

Uncertainty-Aware Hierarchical Factor Graph Optimization for Multi-Camera Extrinsic Calibration of Autonomous Vehicles

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Abstract—Multi-camera perception requires extrinsic calibration that expresses each camera’s position and orientation in a common vehicle frame. Methods that rely on shared target observations or platform motion require overlapping camera views or sufficient maneuvering space, which complicates repeated calibration after sensor replacement or maintenance. Calibration must also account for marker-pose uncertainty that varies with observation conditions. This paper presents an uncertainty-aware hierarchical factor graph framework for calibrating multiple cameras on a stationary vehicle. Stage 1 builds an environmental-marker map from multi-viewpoint observations by a single auxiliary camera and marginalizes the auxiliary-camera poses into a joint prior that preserves correlations among markers. Stage 2 reuses this joint prior when estimating the environmental markers, four wheel markers, and vehicle-camera poses together in a single graph; the wheel markers define the vehicle frame in which the extrinsics are expressed. In both stages, corner-detection uncertainty is propagated to a marker-pose covariance whose inverse serves as an observation-specific information matrix. Camera rigs based on nuPlan, NVIDIA Physical AI (PAI), and an IONIQ vehicle are reproduced in CarMaker and evaluated under three noise settings each. The proposed weighting yields lower rotation, translation, and reprojection RMSEs than the comparison weights in all nine simulation conditions. Across five stationary poses of a real vehicle, the camera-to-camera position and rotation dispersions are 21.0 mm and 0.265° , and the vehicle-frame dispersions are 17.8 mm and 0.323° , all lower than those of the comparison weights. Supplementary material is available at the project page <https://hfgo-hfgo.github.io/hfgo>.

Index Terms—Multi-Camera Extrinsic Calibration, Camera-to-Vehicle Calibration, Reusable Marker Map, Hierarchical Factor Graph Optimization, Uncertainty-Aware Marker Pose Estimation.

I. INTRODUCTION

Mobile robots and autonomous vehicles use multiple cameras with different orientations and fields of view to perceive their surroundings [1]. Fusing image features, objects, and spatial information detected by these cameras requires accurate extrinsic parameters that express each camera’s position and orientation in a common vehicle frame [2]. Inaccurate extrinsics place the same object at different spatial locations across cameras and degrade multicamera fusion. For example, a yaw error of 1° produces a lateral displacement of approximately 0.52 m at a range of 30 m under a simple ray model. Because the sensor mounting can change during production or maintenance, or following an impact or sensor replacement, the extrinsics must be re-estimated before driving. This study addresses such offline calibration before driving; online calibration during operation lies outside its scope.

Among existing offline calibration approaches, motion-based methods require sufficient motion of the sensor platform or calibration target, or an external motion-capture system [3], [4]. Infrastructure-based methods relax field-of-view overlap constraints but require a survey drive to build a map or a calibration facility measured precisely in advance [5], [6]. These requirements entail preparation of workspace, equipment, and facilities that recurs whenever calibration is repeated after sensor replacement or maintenance. When a camera pair does not observe a common target, a shared reference is needed to connect the observations from its cameras. This study uses markers placed in the environment as that shared reference. Because every camera is tied to the environment through the markers it observes, errors in the estimated marker poses propagate to the final extrinsics. Marker-pose uncertainty varies with viewing distance, incidence angle, marker size in the image, and corner-detection quality, and the contribution of each observation should reflect these conditions. A common covariance for every observation [7], [8] or a scalar weight per observation [9] does not represent the variance of each of the six marker-pose degrees of freedom or the correlations between them. The proposed method therefore incorporates the covariance of marker poses estimated from individual images into extrinsic calibration.

An environmental map built once can serve as a common reference for subsequent calibration even when the vehicle or sensor mounting changes. This study stores the marker poses and prior information of a map built without precision surveying, allowing subsequent calibration to omit map construction while the marker arrangement remains unchanged. Because camera poses are estimated in the environmental-map frame, their use on a vehicle also requires the relationship between the environmental and vehicle frames, which this study obtains from markers attached to the four wheels.

This paper presents an uncertainty-aware hierarchical factor-graph framework that connects the cameras of a stationary vehicle to a common environmental map and estimates their extrinsics in the vehicle frame. The framework incorporates image-specific corner-detection uncertainty into both map construction and per-vehicle calibration. Environmental-marker poses estimated with an auxiliary camera and their inter-marker correlations are retained as a joint prior for subsequent calibration. Each vehicle camera connects to the common frame through the mapped markers it observes; camera pairs therefore need not observe the same marker. Fig. 1 shows how observations from the auxiliary and vehicle cameras lead through environmental-

map construction and per-vehicle calibration to the extrinsic parameters.

The contributions of this paper are threefold:

- 1) A hierarchical factor-graph calibration framework that separates environmental-map construction from per-vehicle calibration. A single auxiliary camera builds an environmental-marker map without precision surveying equipment, and a prior retaining inter-marker correlations is reused for subsequent calibration. Each camera is registered to the map without requiring shared marker observations between cameras, target-vehicle motion, or prior pairwise camera calibration, and wheel markers provide the vehicle frame in which the extrinsics are expressed.
- 2) The inverse of the marker-pose covariance propagated from corner-detection uncertainty in a single image serves as the information matrix for each camera-marker factor, accounting for observation-specific six-degree-of-freedom (6DoF) pose uncertainty in both stages.
- 3) An evaluation in simulation and on a real vehicle that quantifies the effect of observation weighting on calibration results and assesses the spread of inter-camera and vehicle-frame extrinsics across five stationary vehicle poses.

II. RELATED WORK

A. Observation Uncertainty in Marker-Based Calibration

Marker-based factor-graph optimization combines marker observations from multiple images into one optimization problem, and its estimates depend on how observation uncertainty is modeled. TagSLAM assigns a common pixel-noise model to AprilTag corner observations and therefore does not explicitly model image-specific corner-detection uncertainty [7]. Reinke *et al.* use a common diagonal covariance for all camera-marker observations, which does not represent changes in marker-pose uncertainty across observation conditions [8]. To distinguish observation confidence, Demidova and Zhuravlev compute scalar weights for AprilTag observations from projected area, viewing angle, and distance from the principal point [9]. Scalar weights, however, cannot directly represent a 6×6 covariance describing uncertainty in each pose degree of freedom and correlations between degrees of freedom. Chen and Schwertfeger construct 6×6 information matrices from the jackknife spread of repeated pairwise sensor calibration results [10]. These matrices weight pairwise sensor calibration results and do not describe the uncertainty of marker observations in individual images.

MarsCoT propagates corner-detection uncertainty through the unscented transform and PnP to obtain a 6×6 marker-pose covariance for a single image [11]. The proposed method uses the inverse of this covariance as the information matrix of each camera-marker factor in both calibration stages.

B. Reusable Calibration Environments

TagSLAM offers a mode for estimating camera position and orientation from a known tag map, but does not address reuse of the same map for extrinsic calibration of multiple vehicles [7]. Heng *et al.* build a natural-feature map using SLAM and SfM on a mapping vehicle, then hold the map fixed for calibrating multiple multicamera vehicles [5]. Their method requires the mapping vehicle to traverse the environment and each target vehicle to drive through the map to collect sufficient 2-D–3-D correspondences. Xie *et al.* use an infrastructure of approximately 500 AprilTags surveyed in advance with a high-resolution 3-D scanner as a common calibration reference. They estimate multisensor extrinsics from observations collected while moving the target vehicle through 30 distinct poses within this infrastructure [6]. The infrastructure-based procedures of Heng *et al.* and Xie *et al.* rely on either a mapping-vehicle survey drive or precision measurement equipment for map construction, and require target-vehicle motion during calibration. The proposed method instead builds an environmental-marker map from multiview auxiliary-camera observations without a survey drive or precision surveying, and reuses its marker poses and joint prior while the marker arrangement remains unchanged.

C. Nonoverlapping Cameras and Vehicle-Frame Calibration

One approach to calibrating nonoverlapping sensors estimates pairwise relative transforms and subsequently combines them. Another approach jointly estimates multisensor extrinsics from common observations or motion constraints. Multi-Sensor Graph-based Calibration (MSG-Cal) and the method of Chen and Schwertfeger combine pairwise sensor calibration results. Graph optimization reduces inconsistencies among those pairwise estimates [10], [12]. MSG-Cal requires each sensor to have at least partial field-of-view overlap with another sensor. The method of Chen and Schwertfeger first calibrates the required sensor pairs. For nonoverlapping camera pairs, it uses a motion-based hand-eye procedure. Zhao *et al.*, by contrast, attach a marker to each target camera and observe the markers with an auxiliary camera. They compose inter-marker transforms along a single path to obtain inter-camera transforms, without global optimization [13]. Other methods exploit motion or external measurements to jointly calibrate multiple sensors without prior pairwise calibration. Multical uses observations collected while translating and rotating the sensor platform for approximately 1–2 minutes. These observations enable calibration of nonoverlapping cameras, LiDAR, and an IMU in one graph [3]. The method of Wang *et al.* constructs inter-sensor constraints using multiple target poses and external motion capture [4]. MC-Calib connects groups of cameras with entirely nonoverlapping fields of view through group motion observed at multiple viewpoints [14].

Expressing calibration results in a vehicle or robot reference frame also requires the relationship between the sensors and that frame. Reinke *et al.* use a CAD-defined reference frame, together with the motion and forward kinematics of

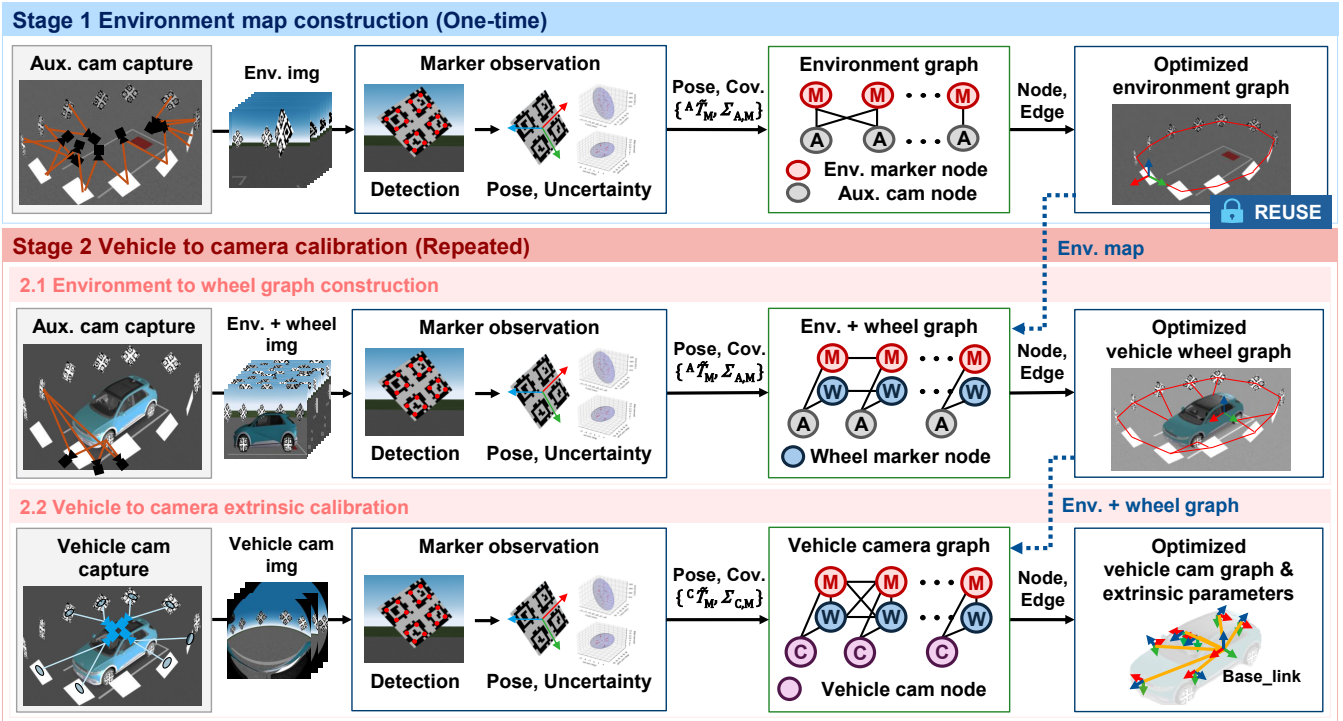


Fig. 1. Overview of the proposed calibration framework. Stage 1 observes environmental markers from multiple viewpoints with an auxiliary camera and marginalizes the auxiliary-camera poses to retain a joint prior over environmental markers. Stage 2 jointly estimates the environmental markers constrained by this prior, the vehicle cameras, and the wheel markers, then derives vehicle-frame camera extrinsics from the wheel-marker-defined vehicle frame. Blue links depict the prior that preserves cross-marker correlations; all environmental markers except the reference marker remain adjustable in Stage 2.

marker-equipped joints, to calibrate multiple cameras in a robot reference frame [8]. The proposed method instead registers both the cameras of a stationary vehicle and the wheel markers that define its frame directly to the Stage 1 environmental map, so it requires no prior pairwise calibration, no common marker observations between camera pairs, and no vehicle motion.

III. PROPOSED METHOD

A. System Overview and Problem Formulation

The method takes, from an auxiliary camera, images of environmental markers for map construction and images of both environmental and wheel markers for vehicle calibration, together with images of environmental markers from the vehicle cameras. Camera intrinsics and projection models, as well as marker identifiers, physical dimensions, and local corner coordinates, are assumed known. The positions and orientations of the environmental markers are not measured in advance. A separate pose node is assigned to each image captured by the moving auxiliary camera. The vehicle remains stationary while images for its calibration are acquired after map construction. Neither overlap between vehicle-camera fields of view nor observation of the same environmental marker by adjacent cameras is required; when such observations occur, both are retained. The estimation graph at each stage must be connected to the frame established by the reference environmental marker, since disconnected camera or marker poses cannot be estimated in this frame.

The coordinate frames and transformations are defined as follows. The reference environmental marker defines the world frame W , and the four wheel markers define the vehicle frame B . The auxiliary-camera frame at capture k and the frame of vehicle camera i are denoted by A_k and C_i , respectively. Environmental- and wheel-marker frames are denoted by M_j^e and M_ℓ^w , respectively; C without a subscript denotes an arbitrary auxiliary- or vehicle-camera frame. For arbitrary frames X and Y , ${}^X\mathbf{T}_Y \in SE(3)$ transforms points from Y to X and represents the pose of Y in X . An observed camera-to-marker relative pose is denoted by ${}^C\tilde{\mathbf{T}}_M$. The primary output is the vehicle-frame extrinsic parameter ${}^B\mathbf{T}_{C_i}$; the inter-camera relative pose ${}^{C_i}\mathbf{T}_{C_j}$ follows from these parameters.

B. Observation-Dependent Marker Pose Reliability

The method estimates both a relative pose and its covariance for each observation to account for reliability that varies with viewing distance, incidence angle, and corner-detection quality. Let $\mathbf{z}_{C,M}$ be the stacked detected corner coordinates, and let $\Sigma_{\mathbf{z},C,M}$ contain the position covariances estimated from local structure tensors around those corners. MarsCoT propagates the corner covariance through an unscented transform and PnP to obtain the marker-pose covariance [11]. This propagation uses the camera intrinsic matrix $\mathbf{K}$, distortion coefficients $\mathbf{D}$, and three-dimensional marker corners $\mathbf{P}_M$. The measured relative pose and information matrix are

defined in (1).

$$\begin{aligned} \left({}^C\tilde{\mathbf{T}}_M, \Sigma_{C,M} \right) &= f_{\text{MarsCoT}}(\mathbf{z}_{C,M}, \Sigma_{z,C,M}; \mathbf{K}, \mathbf{D}, \mathbf{P}_M), \\ \Omega_{C,M} &= \Sigma_{C,M}^{-1}. \end{aligned} \quad (1)$$

Here, ${}^C\tilde{\mathbf{T}}_M$ is the measured marker pose in the camera frame, $\Sigma_{C,M}$ is its 6×6 covariance including correlations among rotational and translational components, and its inverse $\Omega_{C,M}$ serves as the information matrix for that observation. PnP uses the appropriate pinhole or Kannala–Brandt fisheye projection model for each image, and the single-image marker pose is estimated with OpenCV’s `SOLVEPNP_ITERATIVE`.

Optimization compares the relative pose predicted from the current camera and marker poses with the pose measured in the image. Given camera and marker poses ${}^W\mathbf{T}_C$ and ${}^W\mathbf{T}_M$ in the world frame, the predicted pose and its residual relative to the measurement are defined in (2).

$$\begin{aligned} {}^C\hat{\mathbf{T}}_M &= ({}^W\mathbf{T}_C)^{-1} {}^W\mathbf{T}_M, \\ \mathbf{r}_{C,M} &= \text{Log} \left[\left({}^C\tilde{\mathbf{T}}_M \right)^{-1} {}^C\hat{\mathbf{T}}_M \right]^\vee. \end{aligned} \quad (2)$$

The logarithm map converts the pose difference into the 6DoF residual $\mathbf{r}_{C,M}$. The residual and covariance use the same local coordinate frame and component ordering. Given the set $\mathcal{E}$ of camera–marker observations at a stage, optimization minimizes the weighted residual sum in (3).

$$\mathcal{X}^* = \underset{\mathcal{X}}{\text{argmin}} \sum_{e \in \mathcal{E}} \mathbf{r}_e^\top \Omega_e \mathbf{r}_e. \quad (3)$$

Here, $\mathbf{r}_e$ and Ω_e are the residual and information matrix of observation e , so an observation with greater covariance has less influence for a given pose residual.

C. Environmental Map Construction and Reuse

Fig. 2(a) depicts environmental map construction in Stage 1, and Fig. 2(b) and (c), labeled Stage 2.1 and Stage 2.2, depict the connections of wheel markers and vehicle cameras within the single Stage 2 optimization. Environmental marker poses remain the same across vehicles as long as the marker arrangement is unchanged. Wheel-marker and camera poses, in contrast, vary with the vehicle or capture session. Map construction therefore runs once, and its result is passed to subsequent calibrations as a prior. Stage 1 estimates environmental marker poses from multiple auxiliary-camera images. The pose of reference marker M_0^e is fixed to the identity to establish the origin and axes of W . The variables comprise the environmental marker set $\mathcal{M}_e$ and the auxiliary-camera pose set $\mathcal{A}^{(1)}$, with one pose per image. Marginalizing all Stage 1 camera-pose nodes produces a dense Gaussian prior that retains correlations among environmental markers. Because the reference marker is fixed, the prior is defined over the local coordinates of the remaining marker poses. The Stage 1 marker estimates $\mathcal{M}_e^*$ serve as the prior’s linearization point. Partitioning the information matrix of the linearized Stage 1 normal equations

into environmental-marker blocks e and auxiliary-camera blocks A gives the marginalized information matrix in (4).

$$\Lambda_e = \mathbf{H}_{ee} - \mathbf{H}_{eA} \mathbf{H}_{AA}^{-1} \mathbf{H}_{Ae}. \quad (4)$$

The matrix Λ_e preserves joint information across environmental markers. Let $\delta \mathbf{m}$ stack the local displacements from the Stage 1 marker estimates; then $\mathcal{L}_{\text{prior}} = \delta \mathbf{m}^\top \Lambda_e \delta \mathbf{m}$. This term constrains deviations of environmental marker poses from their Stage 1 estimates during Stage 2.

Stage 2 jointly estimates environmental markers $\mathcal{M}_e$, wheel markers $\mathcal{M}_w$, the auxiliary-camera poses $\mathcal{A}^{(2)}$ for the new images, and vehicle-camera poses $\mathcal{C}$. The variables for the two stages are $\mathcal{X}^{(1)} = (\mathcal{M}_e, \mathcal{A}^{(1)})$ and $\mathcal{X}^{(2)} = (\mathcal{M}_e, \mathcal{M}_w, \mathcal{A}^{(2)}, \mathcal{C})$, respectively. Stage 2 minimizes the map-prior cost jointly with the weighted residuals of new observations, as in (5).

$$\begin{aligned} \mathcal{X}^{(2)*} &= \underset{\mathcal{X}^{(2)}}{\text{argmin}} [\mathcal{L}_{\text{prior}}(\mathcal{M}_e) \\ &\quad + \sum_{e \in \mathcal{E}^{(2)}} \mathbf{r}_e^\top \Omega_e \mathbf{r}_e], \\ \text{subject to } {}^W\mathbf{T}_{M_0^e} &= \mathbf{I}. \end{aligned} \quad (5)$$

Stage 1 follows (3), and the Stage 2 observation terms use the same observation-specific information matrices. After Stage 1, the method stores marker poses $\mathcal{M}_e^*$ and their joint information matrix Λ_e ; the original auxiliary-camera poses and observation factors are omitted from subsequent optimizations. The same stored prior is used for every subsequent vehicle, and for every stationary pose of that vehicle, while the marker arrangement remains unchanged. Environmental markers are not held fixed in Stage 2; their adjustment is constrained by the joint Stage 1 information matrix, and wheel-marker and vehicle-camera observations can refine their poses. Wheel markers and vehicle cameras connect to the common environmental map through separate observation paths, so no direct observation between them is required. The term *hierarchical* refers to this sequence of compressing map-construction information into a joint prior and reusing it for vehicle-specific optimization. Under a quadratic approximation at the same linearization point, using this prior is equivalent to marginalizing the Stage 1 camera poses out of a joint optimization over both stages.

Initial estimates propagate from the reference marker or the Stage 1 map poses to other nodes along camera–marker observation paths. Relative poses between markers co-observed in an image are used only for this initialization. Stage 2 includes the map-prior factor and camera–marker factors from new images, avoiding duplicate use of Stage 1 observations.

D. Vehicle-Frame Extrinsic Calibration

The camera poses estimated in Stage 2 are expressed in the world frame W ; using them on the vehicle requires the transformation to the vehicle frame B . The centers of the four wheel markers estimated in Stage 2 define the vehicle frame. The origin is the midpoint of the rear-left and rear-right marker centers, and the unit vector from this

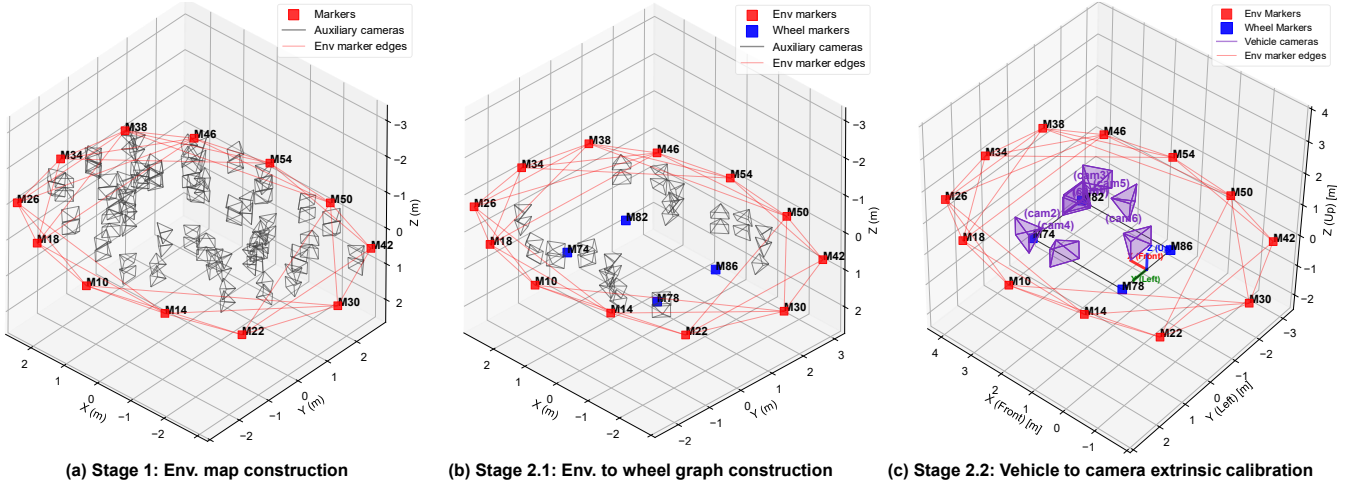


Fig. 2. Optimization graphs for the proposed calibration method. (a) Stage 1 builds the environmental marker map from auxiliary-camera observations. (b) Stage 2.1 connects wheel markers to the map through auxiliary-camera observations of environmental and wheel markers. (c) Stage 2.2 connects vehicle cameras to the map through their environmental-marker observations; the wheel markers define the frame used to report the camera extrinsics. Red and blue squares represent environmental and wheel markers, respectively; gray and purple camera icons represent the auxiliary and vehicle cameras, respectively. Panels (b) and (c) distinguish two observation paths within the same joint Stage 2 optimization graph.

point to the midpoint of the front marker centers defines the forward axis ($+x$). The direction from the rear-right to rear-left marker center is orthogonalized against the forward axis and normalized to define the leftward axis ($+y$); their cross product defines the upward axis ($+z$). These axis directions follow ISO 8855:2011, and ${}^W\mathbf{T}_B$ is computed from the four wheel-marker poses without a separate optimization variable.

The vehicle-frame extrinsic parameter for each vehicle camera and the relative pose between two cameras are computed by (6).

$$\begin{aligned} {}^B\mathbf{T}_{C_i} &= ({}^W\mathbf{T}_B)^{-1} {}^W\mathbf{T}_{C_i}, \\ {}^{C_i}\mathbf{T}_{C_j} &= ({}^B\mathbf{T}_{C_i})^{-1} {}^B\mathbf{T}_{C_j}. \end{aligned} \quad (6)$$

The final calibration depends on the accuracy of the camera and wheel-marker poses as well as the mounting accuracy of the wheel clamps.

IV. EXPERIMENTS

The evaluation examines the effects of observation information matrices and environmental map accuracy on three simulated camera rigs. It then measures the optimization time saved by map reuse, compares the extrinsic error with Chain, a single-path pose composition method, and assesses reprojection fit, cross-pose consistency, extrinsic dispersion, and calibration time on a real vehicle. Fig. 3(a) and (b) show the CarMaker simulation and real-vehicle calibration environments, respectively.

A. Experimental Setup and Evaluation Protocol

The simulation uses camera rigs based on nuPlan with eight cameras, PAI with seven cameras, and IONIQ with ten cameras, reproduced in CarMaker. CarMaker reproduces only the vehicle dimensions and the camera counts, mounting poses, and fields of view; images from the original nuPlan and PAI datasets are not used, so the results do not represent

benchmark performance on those datasets. The simulation provides ground truth (GT) for environmental marker poses and camera extrinsics in the vehicle frame and permits controlled Gaussian noise injection into detected corners. The real-vehicle experiments use four pinhole cameras and four fisheye cameras modeled by Kannala–Brandt on an IONIQ vehicle. The vehicle is placed in five distinct poses, and calibration uses images captured while the vehicle is stationary in each pose. Absolute extrinsic GT is unavailable for the evaluated cameras; thus, the real-vehicle results assess reprojection fit, cross-pose reprojection consistency, and dispersion of the calibrated extrinsics. Both experiments use 2×2 tagStandard41h12 AprilGrids and the AprilTag 3 detector for tag detection. The setup contains 12 environmental marker boards measuring 800×800 mm and four wheel marker boards measuring 500×500 mm. Both experiments use 96 environmental marker images; the simulation uses 32 wheel marker images, and the real-vehicle experiment uses 40. The simulation uses 8, 7, and 10 vehicle-camera images for the nuPlan-based, PAI-based, and IONIQ rigs, respectively, while the real-vehicle experiment uses eight images per vehicle pose.

The information-matrix comparison uses identical camera–marker observations, graph structures, initial estimates, optimizers, and noise realizations; only the observation information matrices and the resulting Stage 1 map priors differ among the methods. The proposed weighting (Ours) uses the inverse of the single-image marker-pose covariance obtained with MarsCoT [11]. Geometry-based weighting (Geom- σ) computes a scalar weight w_g from the projected tag edge lengths, orthogonality of adjacent edges, and distance from the principal point, and uses $\Omega_g = w_g \mathbf{I}_6$ as its information matrix [9]. Fixed weighting (Fixed- σ) applies the same information matrix $\Omega = \mathbf{I}_6$ to every observation. The full joint optimization

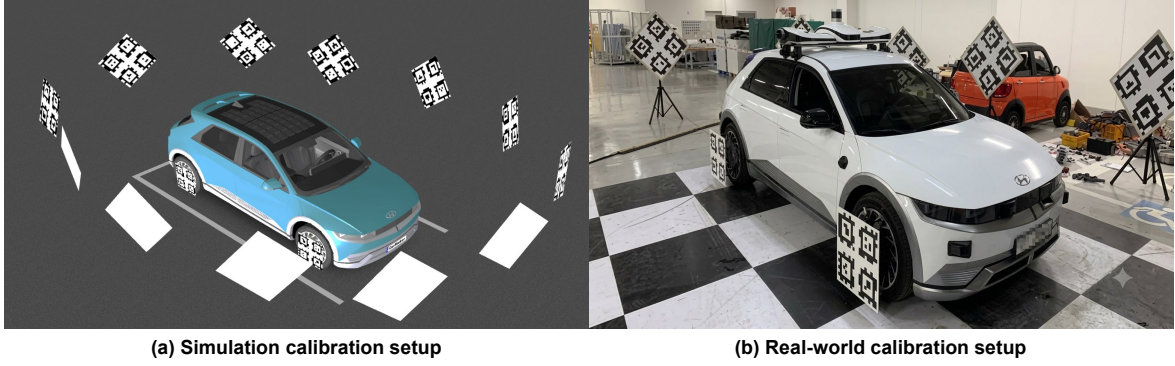


Fig. 3. Calibration environments for (a) the CarMaker simulation and (b) the real IONIQ vehicle. Both contain environmental markers around the vehicle and wheel markers on all four wheels.

baseline, used to assess the separation of map and vehicle states, fixes only the reference environmental marker and jointly optimizes the remaining environmental markers, wheel markers, Stage 1 and Stage 2 auxiliary-camera poses, and vehicle-camera poses. The structural baseline Chain is constructed from the same stationary data using the single-path pose chaining of Zhao *et al.* [13]; it does not reproduce their complete calibration procedure.

Simulation accuracy is evaluated by the rotation and translation RMSE of camera extrinsics relative to GT. Rotation error is the geodesic distance on $SO(3)$, and translation error is the Euclidean distance. Stage 1 map quality is assessed by the rotation and translation RMSE of environmental marker poses relative to GT. Reprojection RMSE is the square root of the mean squared Euclidean distance between detected corners and corners reprojected from the estimated camera and marker poses over all valid corners. The real-vehicle evaluation also reports the standard deviation of reprojection residuals for each vehicle pose. Cross-pose reprojection consistency is assessed by fixing the camera extrinsics estimated at one reference vehicle pose, estimating only the rigid 6DoF transform of the vehicle at each other pose, and then computing reprojection RMSE there. The median RMSE over the 20 ordered pairs from five poses serves as the consistency metric. Cross-pose extrinsic dispersion is computed from relative camera poses and vehicle-frame camera poses estimated at the five vehicle poses. Position dispersion is the square root of the sum of the three sample variances; rotation dispersion is $\sqrt{\sum_k \theta_k^2 / (n-1)}$, where θ_k is the geodesic angle of pose k from the chordal mean and $n = 5$. Camera-to-camera values are averaged over 28 pairs, and vehicle-to-camera values are averaged over eight cameras. Runtime includes only subsequent optimization after Stage 1 map construction and prior computation. Graph optimization uses the Levenberg–Marquardt method in GTSAM. In tables comparing the three weighting methods, bold denotes the minimum and underlining the second-smallest value.

B. Effects of Observation Information Matrices

Each simulated rig is evaluated under three corner-noise settings for environmental marker images and wheel-

marker/vehicle-camera images: none/none, none/1 px, and 1 px/1 px, yielding the nine conditions summarized in Table I.

The proposed weighting achieves the lowest rotation, translation, and reprojection RMSE in all nine conditions in Table I. With Gaussian noise of 1 px standard deviation applied to all detected marker corners, the mean rotation, translation, and reprojection RMSE across the three rigs are 0.614° , 38.5 mm, and 1.474 px, respectively. Without additional corner-noise injection, all three methods have translation errors of a few millimeters; with 1 px noise, their errors increase, and the gap between the proposed and comparison weights widens.

C. Environmental Map Quality and Reuse Efficiency

Table II compares the Stage 1 environmental marker poses estimated from multiview auxiliary-camera observations with simulation GT.

Table II shows that, without additional corner-noise injection, Stage 1 map rotation and translation RMSE across the three rigs range from 0.003° to 0.007° and from 0.219 to 0.398 mm, respectively. With 1 px standard-deviation Gaussian noise applied to environmental marker corners, the ranges become 0.187° – 0.252° and 7.790–13.295 mm. Under the evaluated conditions, the Stage 1 map poses are therefore adequate linearization points for the Stage 2 prior. Marginalizing the Stage 1 auxiliary-camera poses removes the need to reprocess the original images in subsequent Stage 2 calibrations, and calibration with the joint prior matches full joint optimization to the reported decimal places. For simulated IONIQ, Stage 2 optimization takes 2.822 s, 61.4% less than the 7.311 s required by full joint optimization. For the real-vehicle configuration, the Stage 2 graph contains 64 pose nodes compared with 160 in full joint optimization, including the reference marker.

D. Calibration of Camera Pairs Without Common Marker Observations

Each simulated rig includes camera pairs without common environmental marker observations. The proposed method connects each such camera to the map through its own

TABLE I

CALIBRATION ERRORS IN CARMAKER SIMULATION FOR DIFFERENT INFORMATION MATRICES, WITH THE STAGE 1 JOINT PRIOR PASSED TO STAGE 2. EACH CELL REPORTS ROTATION RMSE [DEG] / TRANSLATION RMSE [MM] OF VEHICLE-FRAME CAMERA EXTRINSICS / REPROJECTION RMSE [PX] OF VEHICLE-CAMERA OBSERVATIONS. NOISE SETTINGS LIST ENVIRONMENTAL-MARKER IMAGES BEFORE WHEEL-MARKER AND VEHICLE-CAMERA IMAGES; “NONE” MEANS NO INJECTED CORNER NOISE.

Rig	Noise (env. / wheel, vehicle)	Ours	Geom- σ	Fixed- σ
nuPlan	None / None	0.017 / 1.343 / 0.0399	<u>0.079 / 3.073 / 0.1241</u>	0.101 / 3.490 / 0.1365
nuPlan	None / 1 px	0.673 / 41.008 / 1.8295	<u>1.846 / 91.550 / 5.2363</u>	2.389 / 127.178 / 6.4214
nuPlan	1 px / 1 px	0.613 / 38.286 / 1.7588	<u>1.849 / 107.212 / 8.4029</u>	2.166 / 132.220 / 9.0703
PAI	None / None	0.019 / 1.512 / 0.1337	<u>0.070 / 3.768 / 0.3496</u>	0.076 / 4.038 / 0.3024
PAI	None / 1 px	0.518 / 25.979 / 1.2495	<u>2.358 / 137.768 / 5.4391</u>	<u>1.989 / 147.933 / 6.0149</u>
PAI	1 px / 1 px	0.449 / 27.079 / 1.2917	<u>2.478 / 132.272 / 5.4777</u>	<u>2.157 / 141.676 / 5.7237</u>
IONIQ	None / None	0.032 / 1.502 / 0.1539	<u>0.075 / 4.453 / 0.3372</u>	0.071 / 5.112 / 0.3637
IONIQ	None / 1 px	0.549 / 38.241 / 1.3185	<u>1.938 / 124.200 / 5.6532</u>	3.356 / 215.497 / 7.7577
IONIQ	1 px / 1 px	0.779 / 50.035 / 1.3710	<u>1.786 / 112.082 / 6.3657</u>	3.112 / 200.614 / 7.5469

TABLE II

ROTATION RMSE [DEG] AND TRANSLATION RMSE [MM] OF STAGE 1 ENVIRONMENTAL MARKER POSES RELATIVE TO SIMULATION GT.

Rig	Env. corner noise	Rotation ↓	Translation ↓
nuPlan	None	0.007	0.390
nuPlan	1 px	0.210	10.293
PAI	None	0.006	0.398
PAI	1 px	0.187	7.790
IONIQ	None	0.003	0.219
IONIQ	1 px	0.252	13.295

TABLE III

VEHICLE-CAMERA EXTRINSIC ERRORS (ROTATION RMSE [DEG] / TRANSLATION RMSE [MM]) FOR THE PROPOSED METHOD AND SINGLE-PATH POSE COMPOSITION (CHAIN) UNDER 1 PX CORNER NOISE.

Rig	Ours	Chain
nuPlan	0.613 / 38.286	6.627 / 264.728
PAI	0.449 / 27.079	4.960 / 199.033
IONIQ	0.779 / 50.035	2.031 / 76.557

environmental-marker observations, and connects the map to the vehicle frame through the Stage 1 prior and the auxiliary-camera observations of wheel markers, without moving the vehicle or providing pairwise camera extrinsics. The rotation and translation RMSE in Table I therefore also quantify vehicle-frame calibration under these conditions. Table III compares the proposed method with Chain on the same stationary data, with 1 px standard-deviation Gaussian noise applied to all detected marker corners. The proposed method has lower rotation and translation RMSE on all three rigs, with reductions of 61.6–90.9% for rotation and 34.6–86.4% for translation.

E. Real-Vehicle Reprojection, Extrinsic Dispersion, and Subsequent Calibration Time

Table IV summarizes the real-vehicle reprojection errors at each of the five stationary poses and the cross-pose reprojection consistency.

The proposed weighting achieves the lowest reprojection RMSE and residual standard deviation at every pose in

TABLE IV

REPROJECTION ERRORS [PX] FOR EIGHT REAL-VEHICLE CAMERAS: (A) RMSE / RESIDUAL SD OVER ALL VALID CORNERS AT EACH POSE; (B) MEDIAN CROSS-POSE RMSE OVER 20 ORDERED POSE PAIRS.

(a) RMSE / residual SD by pose			
Pose	Ours	Geom- σ	Fixed- σ
Pose1	0.832 / 0.455	1.693 / 0.923	1.344 / 0.657
Pose2	0.611 / 0.299	<u>1.323 / 0.702</u>	1.433 / 0.870
Pose3	0.645 / 0.366	1.395 / 0.774	1.188 / 0.641
Pose4	0.657 / 0.354	1.445 / 0.872	<u>1.173 / 0.677</u>
Pose5	0.860 / 0.501	1.564 / 0.904	<u>1.256 / 0.703</u>
(b) Median cross-pose reprojection RMSE			
Ours	Geom- σ	Fixed- σ	
1.35	2.11	<u>2.02</u>	

TABLE V

CROSS-POSE DISPERSION OF EXTRINSICS OVER FIVE STATIONARY REAL-VEHICLE POSES, AVERAGED OVER 28 CAMERA PAIRS (CAMERA-TO-CAMERA) AND EIGHT CAMERAS (VEHICLE-TO-CAMERA); SMALLER VALUES INDICATE GREATER REPEATABILITY.

Method	Camera-to-camera		Vehicle-to-camera	
	Position [mm]	Rotation [°]	Position [mm]	Rotation [°]
Ours	21.0	0.265	17.8	0.323
Geom- σ	22.3	0.308	18.9	0.469
Fixed- σ	22.9	0.317	19.1	0.446

Table IV(a). The median cross-pose reprojection RMSE in Table IV(b) is also lower for the proposed weighting at 1.35 px, compared with 2.11 px for Geom- σ and 2.02 px for Fixed- σ . The proposed weighting therefore improves both per-pose reprojection fit and cross-pose consistency on real images. Table V summarizes cross-pose extrinsic dispersion.

In Table V, the proposed method has camera-to-camera position and rotation dispersions of 21.0 mm and 0.265° and vehicle-to-camera dispersions of 17.8 mm and 0.323°. All four values are lower than those of the comparison weights. These values indicate repeatability across the five stationary poses, not accuracy relative to absolute extrinsic GT. Vehicle-to-camera dispersion also includes cross-pose variation in the

vehicle frame defined by the wheel markers. For the real IONIQ, Stage 2 optimization takes 5.720 s, 63.2% less than the 15.541 s required by full joint optimization.

F. Discussion and Limitations

The environmental map prior can be reused only while the marker arrangement remains unchanged; changes to that arrangement require reconstruction. Stage 1 map errors can affect subsequent calibration even when Stage 2 readjusts environmental marker poses. The method accounts for single-image corner-detection uncertainty but does not separately model marker-printing tolerances, board flatness, residual camera-intrinsic errors, or wheel-clamp installation errors. The statistical consistency of the optimized state covariance has not been evaluated. Because absolute GT is unavailable for the real environmental map, quantitative map-quality evaluation is limited to simulation.

V. CONCLUSION

This paper presented an uncertainty-aware hierarchical factor-graph framework that calibrates multiple cameras on a stationary vehicle in the vehicle frame, including camera pairs without common environmental marker observations. An environmental marker map built with one auxiliary camera and no precision surveying equipment is compressed into a joint prior and reused for vehicle-specific calibration without reprocessing the original images. Four wheel markers define the vehicle frame, so neither vehicle motion nor prior pairwise camera calibration is required. Both stages use the inverse of the marker-pose covariance propagated from corner-detection uncertainty as each observation's information matrix. In all nine CarMaker simulation noise conditions, this weighting achieves lower rotation, translation, and reprojection RMSE than the comparison weights, and on the real vehicle it yields lower median cross-pose reprojection RMSE and lower camera-to-camera and vehicle-to-camera dispersion across five stationary poses. Relative to single-path relative-pose composition, it reduces vehicle-frame rotation RMSE by 61.6–90.9% and translation RMSE by 34.6–86.4% under 1 px corner noise.

Future work will assess absolute extrinsic errors for real-vehicle camera pairs without common environmental marker observations using independent precision measurements, map-reuse accuracy across vehicles, and calibration repeatability after wheel-clamp reinstallation.

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